

To-Do Date: May 27 at 11:59pm

Setting
the
Stage

The Turn of the Century and the 2nd Industrial Revolution

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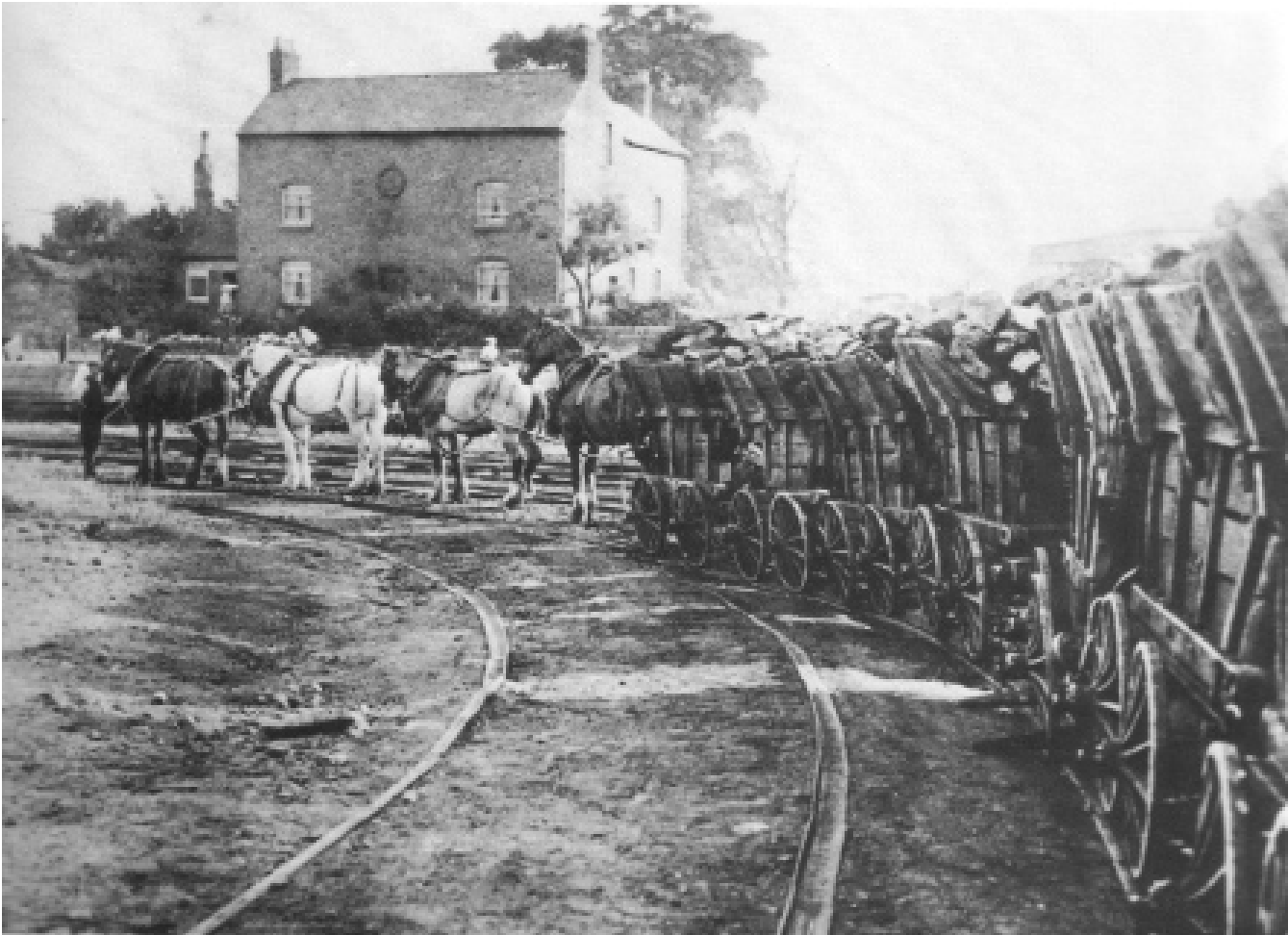


Between 1820 and 1860, the visual map of the United States was transformed by unprecedented urbanization and rapid territorial expansion. These changes mutually fueled the Second Industrial Revolution which peaked between 1870 and 1914. Between the annexation of Texas (1845), the British retreat from Oregon country, and The Treaty of Guadalupe-Hidalgo (1848) which cemented Mexican cession of the Southwest to the United States, territorial expansion exponentially

rewrote the competing visions that free-soilers, European immigrants, industrial capitalists, and Native Americans held for the future of the American Empire.

"If a Western Rip Van Winkle had fallen asleep in 1869 and awakened in 1896, he would not have recognized the lands that the railroads had touched. Bison had yielded to cattle; mountains had been blasted and bored. Great swaths of land that had once whispered grass now screamed corn and wheat. Nation-states had conquered Indian peoples, slaughtering some of them and confining and controlling most of them. Population had increased across much of this vast region, and there were growing cities along its edges. A land that had once run largely north-south now ran east-west. Each change could have been traced back to the railroads." - Richard White, "Railroaded"

Need for Railroads



📷 “Illustrated History of the Railroads” Benjamin Outram’s Little Eaton Gangway in July 1908.

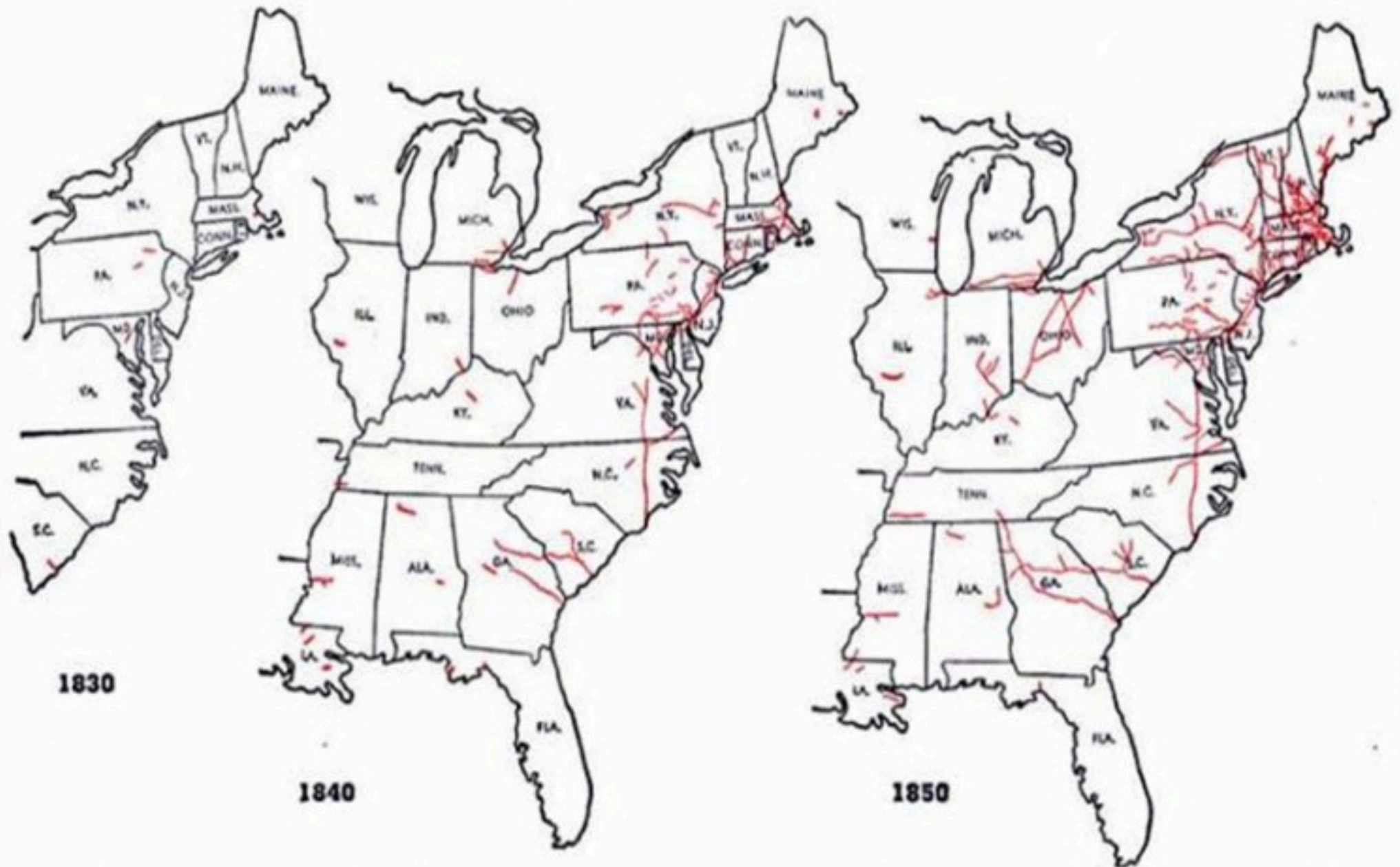
The need for massive industry was obvious: in order to reach California’s burgeoning port cities like San Francisco and to expedite the extraction of gold from the mines, railroad tracks would need to be laid across the plains to reach the Pacific and open up trade networks. Questions abounded about the character this new American territory would take: would it be reliant on slave labor and fulfill Jefferson’s original vision of an agrarian republic? Would corporations or the federal government lay down the required infrastructure to ‘tame the West’? Still, others wondered if turning over the bison laden Plains to New York-based corporations would stifle the American dream for America’s second and third sons. Still, others

believed the technological innovations of the Second Industrial Revolution were the unstoppable culmination of modern civilization propelling the fulfillment of Manifest Destiny. Questions of this nature were not new in American history. Throughout the first half of the nineteenth century, Americans were forced to adjust to the implications of the First Industrial Revolution. 1750 to 1850 marked a century of heightened industrial activity centered around textiles. After the invention of steam power and the cotton gin by Eli Whitney in 1793, cotton could be shipped from the American South by New England ships to the vast textile factories of Great Britain, producing a reverse triangle trade around a single global commodity. These developments were hailed by some as “progress,” but the pace, scale, and reliance on slave labor on these developments instilled in others a great sense of anxiety and fear.

Although the economic and social problems of the first Industrial Revolution distressed many, [these concerns were set aside during the nation's bloody Civil War](https://ushistoryscene.com/article/civil-war-west-expansion/) (1861-1865).

- [Chinese Railroad Workers and the Golden Spike](https://ushistoryscene.com/article/chinese-railroad-workers-and-the-golden-spike/)

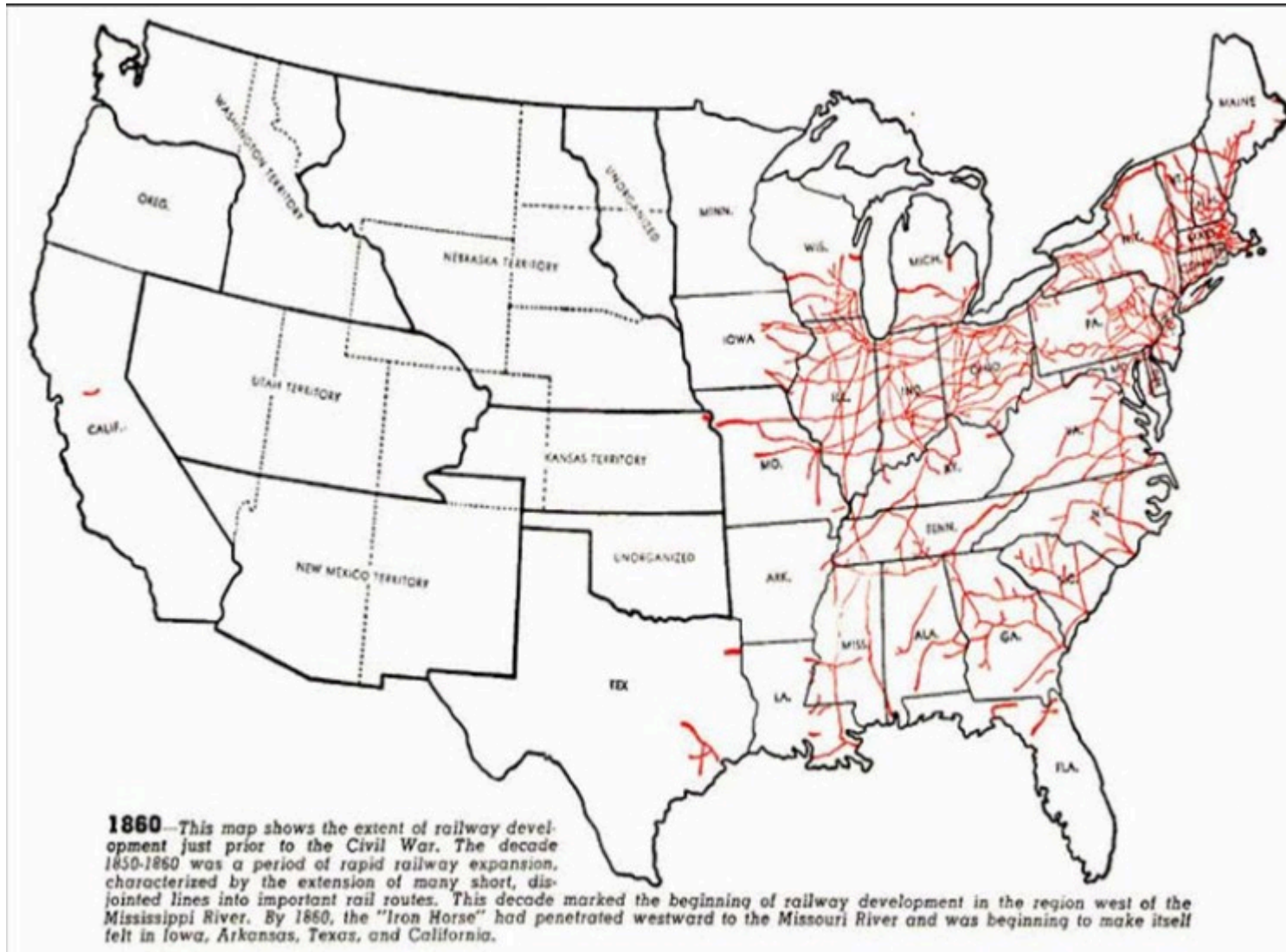
The following maps demonstrate the advancement of the railroad before the Civil War (as always, click to enlarge image):



The early stages of railway development in America are shown by this set of maps. During the decade 1830-1840, the total length of completed railroad lines increased from 23 to 2,808 miles, and during the next ten years, more than 6,200 miles of railroad were opened, bringing the total network up to 9,021 miles in 1850. The most intensive growth during this period was in the Atlantic and Seaboard states. In 1850, a trip from Boston or New York to Chicago was made by rail and lake steamers or by stagecoaches and required several days. One could travel all the way from Boston to Philadelphia, New

...of by rail, and requires several days. One could travel all the way from Boston to Wilmington, North Carolina, by rail, with several changes of cars and a few ferry trips en route. During the first twenty years of railway development, covered by these maps, the population of the United States nearly doubled.

U.S. Railroad Map, 19th Century.



U.S. Railroad Map, 1860.

American Economic Growth

In the aftermath of the Civil War and Reconstruction, the American economy grew considerably as it entered “The Second Industrial Revolution,” generally recognized as the period between 1870 and 1914. The U.S. was awash in an abundance of natural resources from its newly acquired territories, a growing supply of labor immigrating from Europe, and the migration of emancipated African Americans North and West, an expanding market for manufactured goods, and the availability of capital for investment.

The Second Industrial Revolution took local communities and their new products out of the shadow of large regional agricultural based economies which was assisted by new labor forces and production techniques. During the Second Industrial Revolution, innovations in transportation, such as roads, steamboats, the Erie Canal, and most notably railroads, linked distant, previously isolated communities together.

Transporting Products

For the first time, goods from the American interior could be shipped directly to the Atlantic, and vice versa. Being able to ship products great distances transformed the nature of economic activity in the United States. Before the development of this elaborate transportation and communication system, economies were localized and often based on a barter system. The transportation revolution opened up new markets for farmers, industrialists, and bankers who could now bring crops cotton in the Mississippi River Valley, wheat in the Midwest, and manufactured goods in upstate New York into a global market based on credit. Similarly, the expansion of the railroad brought a dramatic reduction in the time and money it took to move heavy goods, creating new opportunities for wealth at a time when two-thirds of all Americans still resided on farms.

Government Involvement

The federal government actively participated in this growth by promoting industrial and agricultural development. High tariffs were enacted to protect American industry from foreign competition, land was granted to railroad companies to

encourage construction, and the army was employed to forcibly remove Indians from western land desired by farmers and mining companies. The rapid growth of factory production, mining, and railroad construction all boosted the new industrial economy and stood in stark contrast to the previous small farm and artisan workshop economy of the pre-Civil War era.

By 1913, the United States produced one-third of the world's industrial output—more than the total of Great Britain, France, and Germany combined. The living standards and the purchasing power of money increased rapidly, as new technologies played an ever-increasing role in the daily lives of working- and middle-class citizens. Between 1870 and 1920, almost 11 million Americans moved from farm to city, and another 25 million immigrants arrived from overseas. By 1920, for the first time in American history, the census revealed more people lived in cities than on farms.

Interconnected Growth





📷 **Puck Magazine (September 7, 1904)** This cartoon portrays a “Standard Oil” storage tank as an octopus with tentacles wrapped around the steel, copper, and shipping industries, as well as a statehouse, the U.S. Capitol, and one tentacle reaching for the White House.

Inventions during the Second Industrial Revolution were interconnected. The railroad spurred the growth of the telegraph machine. Telegraph lines and railroad lines inextricably bound together as telegraph polls dotted the distance of railroad lines. The telegraph, and later the telephone, ushered in the era of instant communication and brought about, in the words of cultural historian Stephen Kern, “the annihilation of distance.” This was a profound change for Americans. The “local” shot outward to the “national” and even “international” as a new sense of world unity was established through these new technologies. These technologies also increased the pace of life and the manner in which people worked and lived.

Major Technological Advances of the Second Industrial Revolution

- **1870s.** Automatic signals, air brakes, and knuckle couplers on the railroads; the Bessemer and then the open-hearth process in the steel mills; the telephone, electric light, and typewriter.
- **1880s.** The elevator and structural steel for buildings, leading to the first “skyscrapers.”
- **1890s.** The phonograph and motion pictures; the electric generator, contributing to modern household items such as refrigerators and washing machines and gradually replaced water and steam-powered engines; and the internal combustion engine, which made possible the first automobiles and the first airplane flight by the Wright brothers in

1903.

Unstable Growth

The economic growth during this time period was extraordinary but unstable. The world economy experienced harsh depressions in 1873 and again in 1897. Businesses competed intensely with each other and corporations battled to gain control of industries. Countless companies failed and others were bought up by larger corporations which eventually ruled the marketplace.

For those who were able to capitalize on these technological advancements, the Second Industrial Revolution was highly profitable. During the Depression of 1873, the soon-to-be industry giant, Andrew Carnegie established a steel company which controlled every phase of business from raw materials to transportation, manufacturing, and distribution.

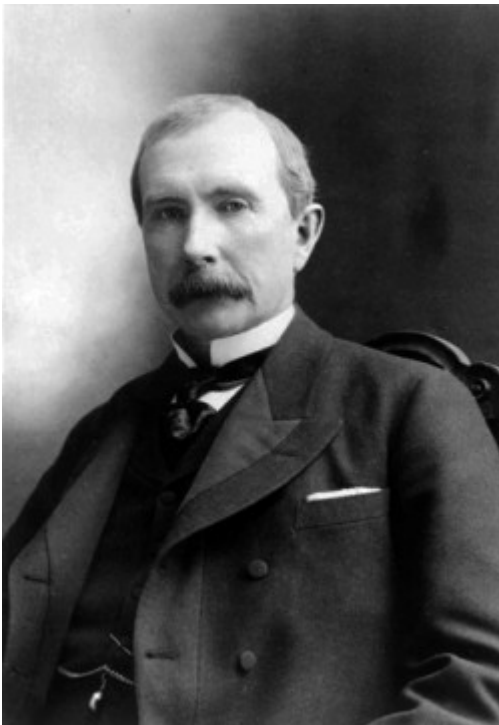


Andrew Carnegie

Carnegie Fortune

By the 1890s, Carnegie dominated the steel industry and had accumulated a fortune worth millions. His steel factories were the most technologically advanced in the world, although this honor came at a price for his workers. Carnegie ran his companies with a dictatorial hand; his factories operated around the clock and workers were burdened with long hours. Yet, at the same time, Carnegie believed that the rich had a moral obligation to promote the advancement of society and he distributed much of his wealth to various philanthropies, especially towards the creation of public libraries throughout the country.

Like Carnegie, John D. Rockefeller also accumulated enormous amounts of wealth, although his money came through domination of the oil industry.



John D. Rockefeller (1885)

Beating the Competition

Rockefeller annihilated rival oil firms through committed competition, secret deals with railroad companies, fixed prices

Rockefeller annihilated rival oil firms through committed competition, secret deals with railroad companies, fixed prices, and production quotas. He bought out competing oil refineries and managed all aspects of the operation, including drilling, refining, storage, and distribution. Before long, Rockefeller's Standard Oil Company controlled a majority of the nation's oil industry. Like Carnegie, Rockefeller publicly supported a number of philanthropies, yet privately domineered over his workers and bitterly fought their efforts to organize and unionize.

Conclusion

The Second Industrial Revolution fueled the Gilded Age, a period of great extremes: great wealth and widespread poverty, great expansion and deep depression, new opportunities, and greater standardization. Economic insecurity became a basic way of life as the depressions of the 1870s and 1890s put millions out of work or reduced pay. Those who remained in the industrial line of work experienced extremely dangerous working conditions, long hours, no compensation for injuries, no pensions, and low wages. But for a limited minority of workers, the industrial system established new forms of freedom. Skilled workers received high wages in industrial work and oversaw a great deal of the production process. Economic independence now required a technical skill rather than ownership of one's own shop and tools. It was labeled "progress" by its proponents, but those who worked the floor at the factory knew it came at a price.